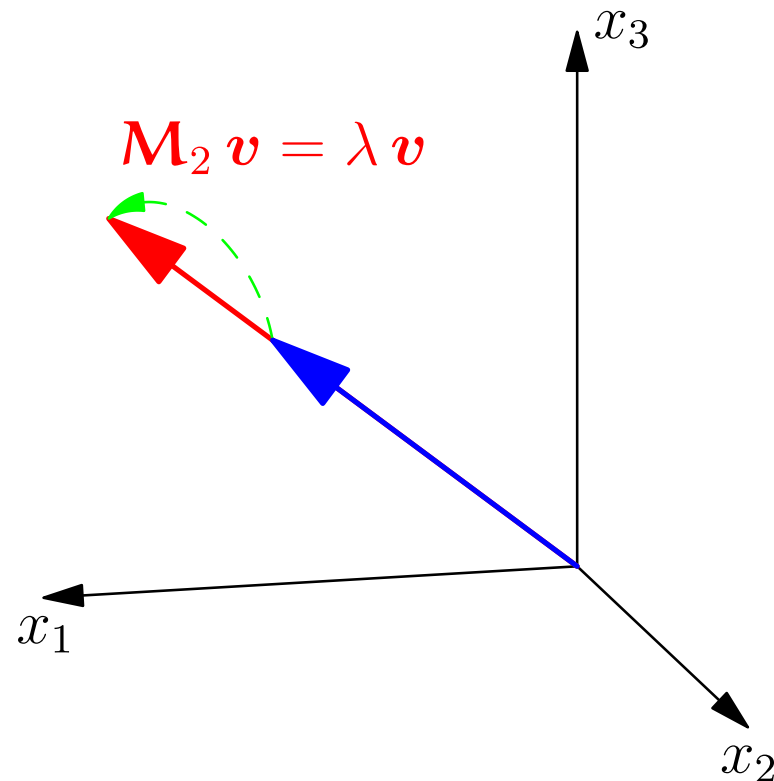


# Mathematics of Machine Learning

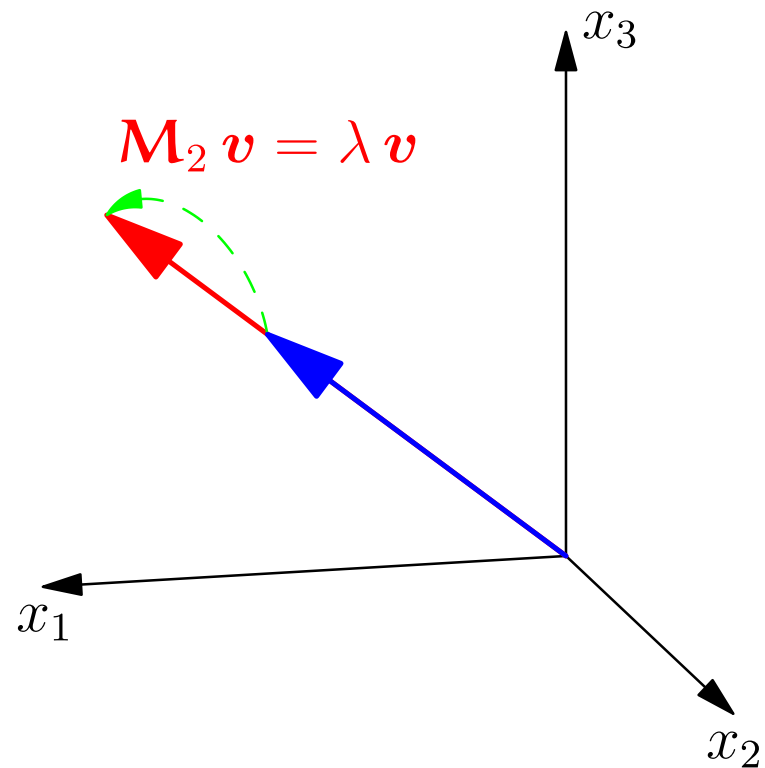
## *Eigensystems*



*Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank*

# Outline

1. **Eigenvectors**
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation



# Eigenvector equation

- Eigen-systems help us to understand mappings
- A vector  $v$  is said to be an **eigenvector** if

$$Mv = \lambda v$$

- $M$  is square (i.e.  $n \times n$ )
- Where the number  $\lambda$  is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

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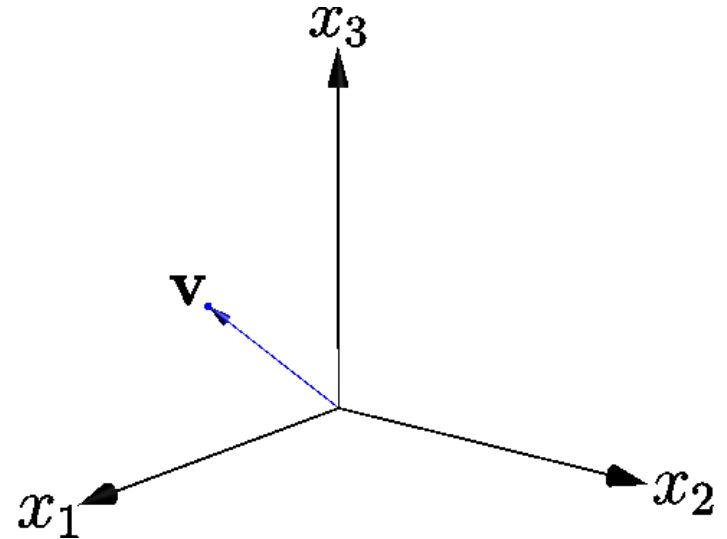
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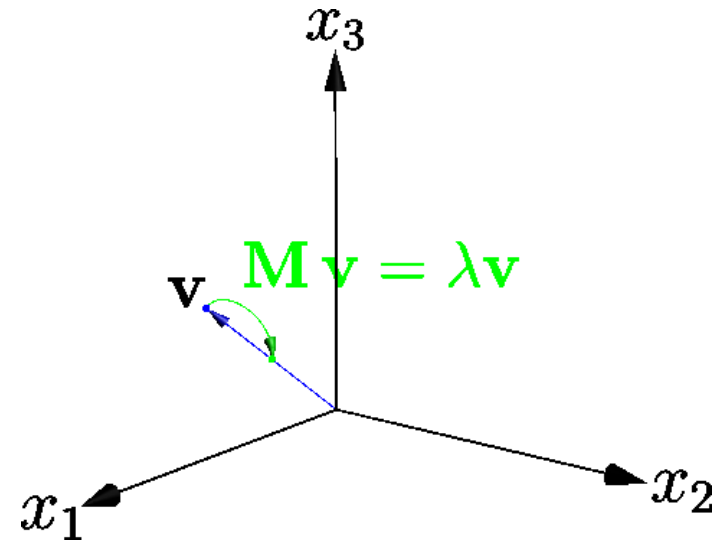


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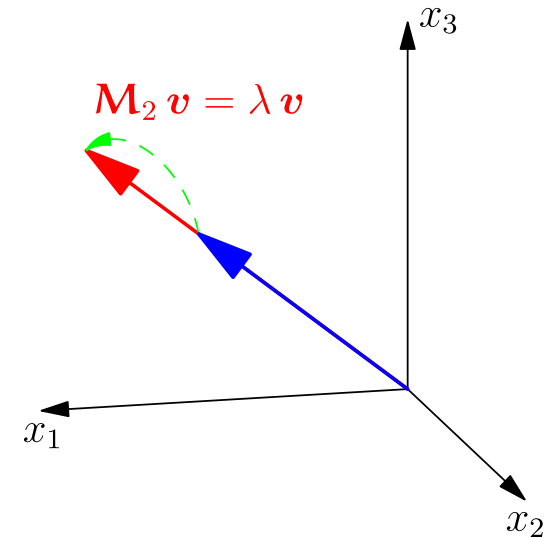


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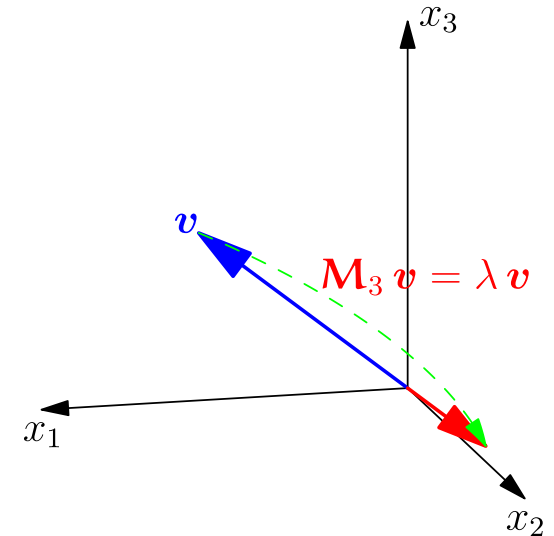


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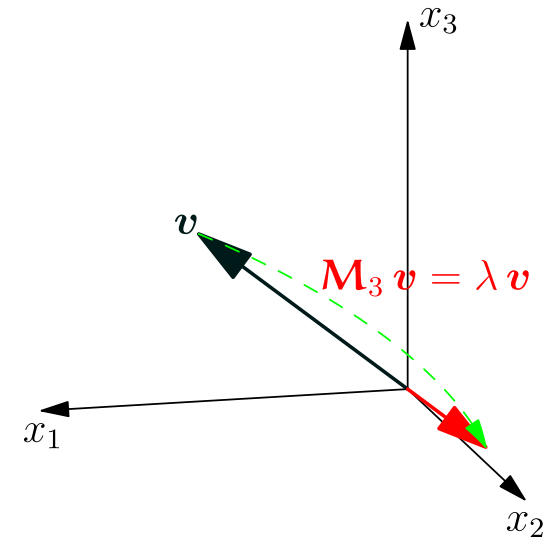


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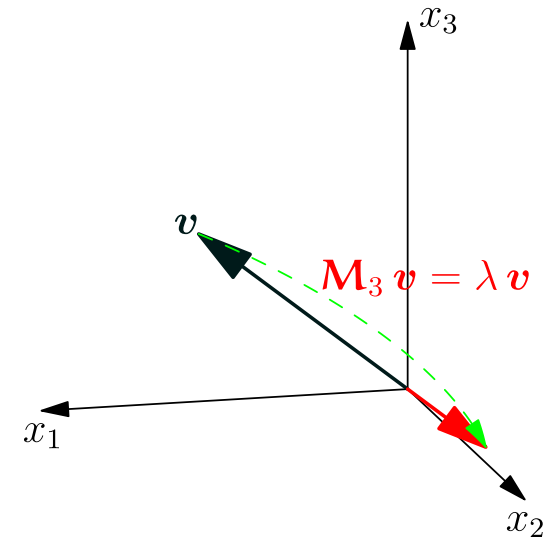


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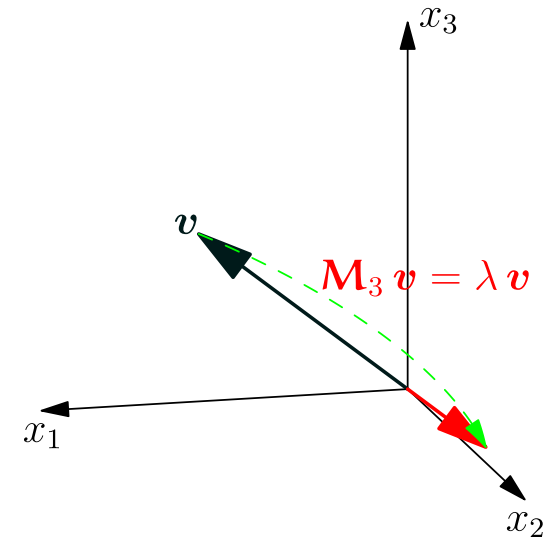


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# Symmetric Matrices

- If  $\mathbf{M}$  is an  $n \times n$  **symmetric** matrix then it has  $n$  real orthogonal eigenvectors with real eigenvalues
- We denote the  $i^{th}$  eigenvector by  $\mathbf{v}_i$  and the corresponding eigenvalue by  $\lambda_i$  so that

$$\mathbf{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

- Orthogonal means that if  $i \neq j$  then

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# Proof of Orthogonality

- $(\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i)^\top$  implies  $\mathbf{v}_i^\top \mathbf{M}^\top = \lambda_i\mathbf{v}_i^\top$
- When  $\mathbf{M}$  is symmetric then  $\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i \Rightarrow \mathbf{v}_i^\top \mathbf{M} = \lambda_i\mathbf{v}_i^\top$
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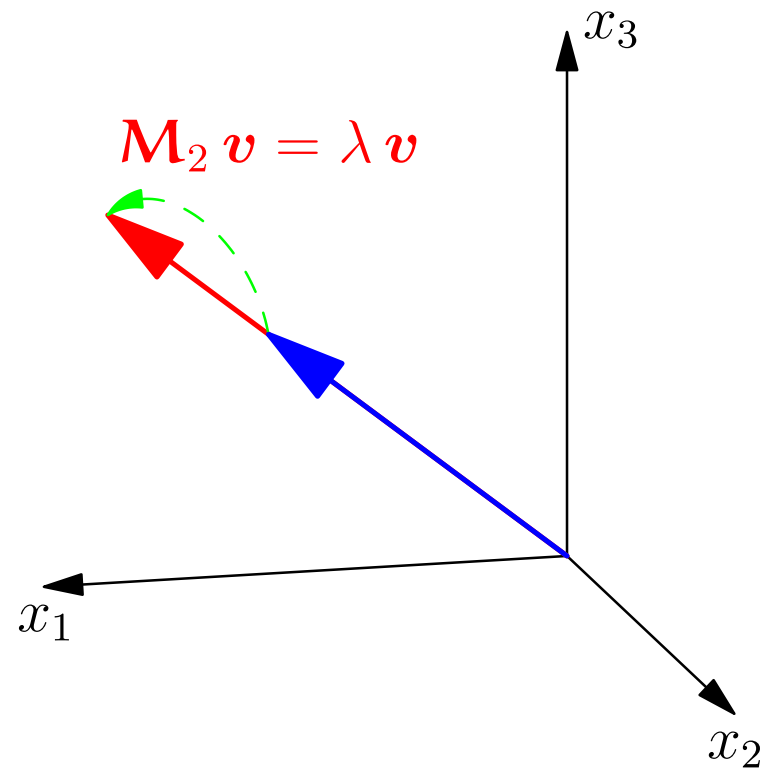
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# Outline

1. Eigenvectors
2. **Orthogonal Matrices**
3. Eigen Decomposition
4. Low Rank Approximation



# Orthogonal Matrices

- We can construct an **orthogonal** matrix **V** from the eigenvectors

$$\mathbf{V} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$$

- Matrix **V** is an  $n \times n$  matrix
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- We have shown that  $\mathbf{V}^T \mathbf{V} = \mathbf{I}$
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- A matrix,  $\mathbf{M}$ , will be singular (uninvertible) if there exists a vector  $\mathbf{x}$  ( $\neq \mathbf{0}$ ) such that

$$\mathbf{M}\mathbf{x} = \mathbf{0}$$

- Now if there exists such a vector such that  $\mathbf{V}\mathbf{x} = \mathbf{0}$  then multiply by  $\mathbf{V}^T$  we get

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- Thus  $\mathbf{V}$  is invertible

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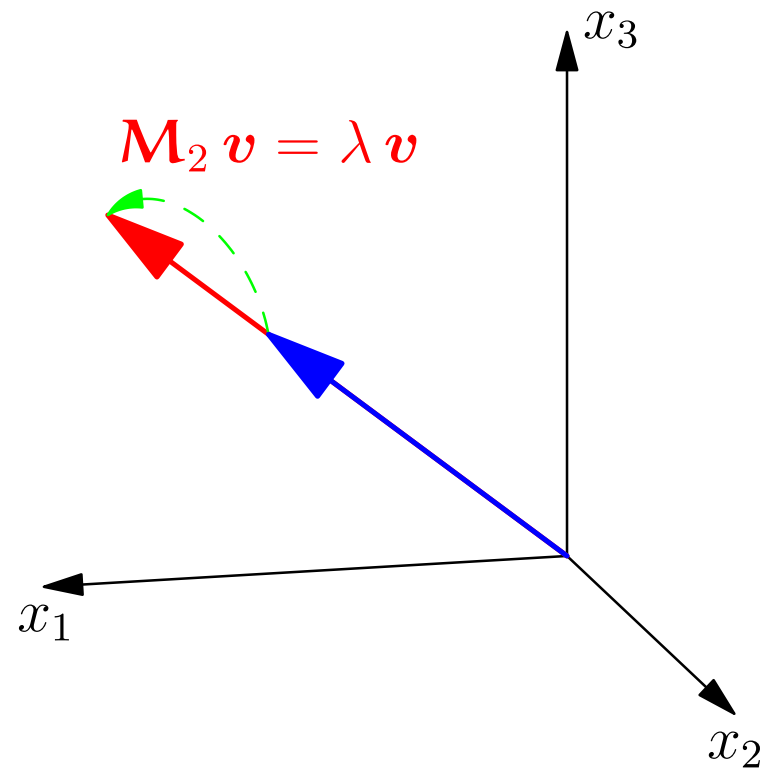
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# Outline

1. Eigenvectors
2. Orthogonal Matrices
3. **Eigen Decomposition**
4. Low Rank Approximation



# Matrix Decomposition

- Taking the matrix of eigenvectors,  $V$ , then

$$MV = M(v_1, v_2, \dots, v_n) = (\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n) = V\Lambda$$

- where  $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$

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- Very important *similarity transform*

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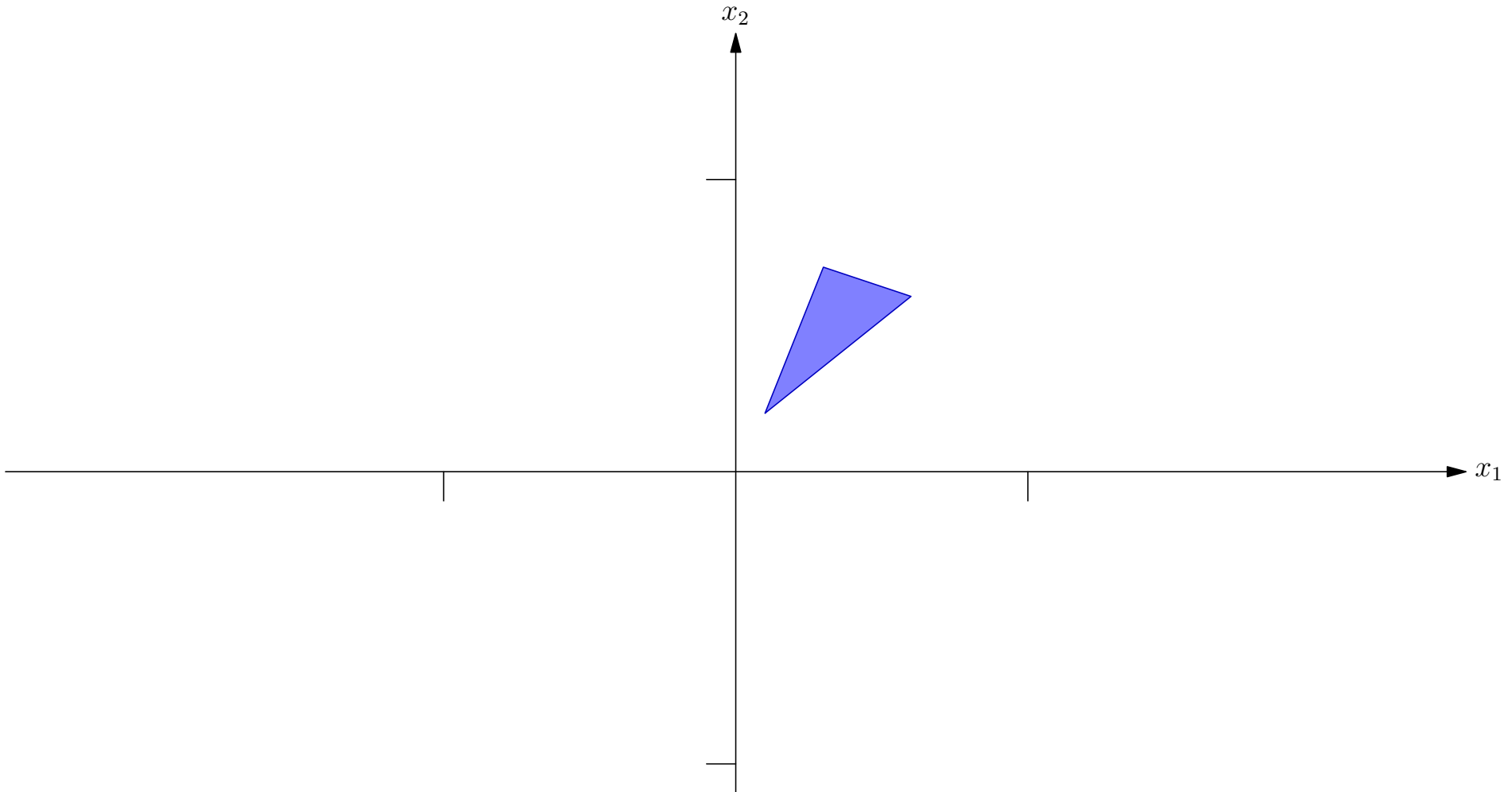
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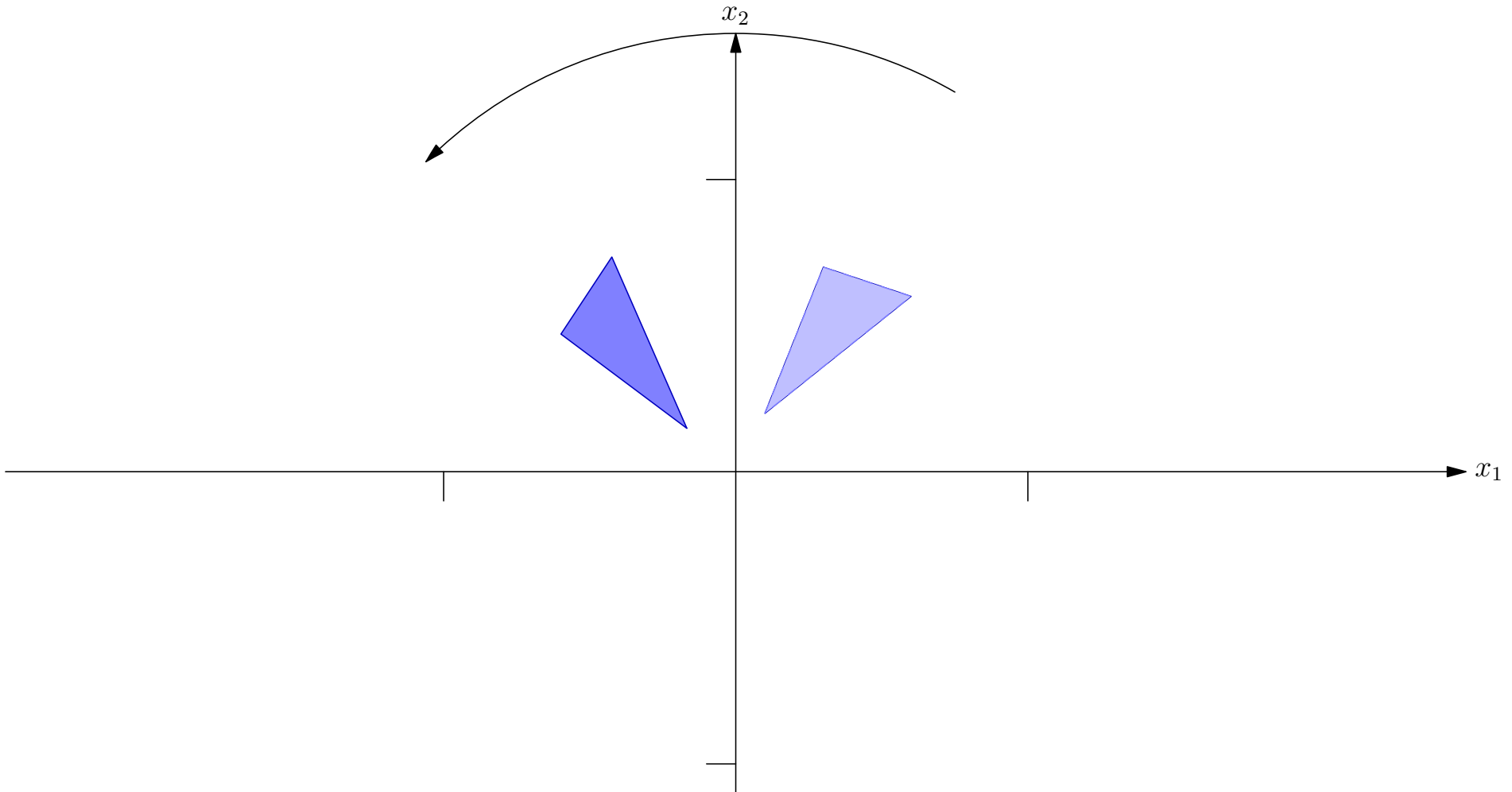
# Mappings by Symmetric Matrices

$$\mathbf{M} = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \begin{pmatrix} \cos(-75) & \sin(-75) \\ -\sin(-75) & \cos(-75) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75) & \sin(75) \\ -\sin(75) & \cos(75) \end{pmatrix}$$



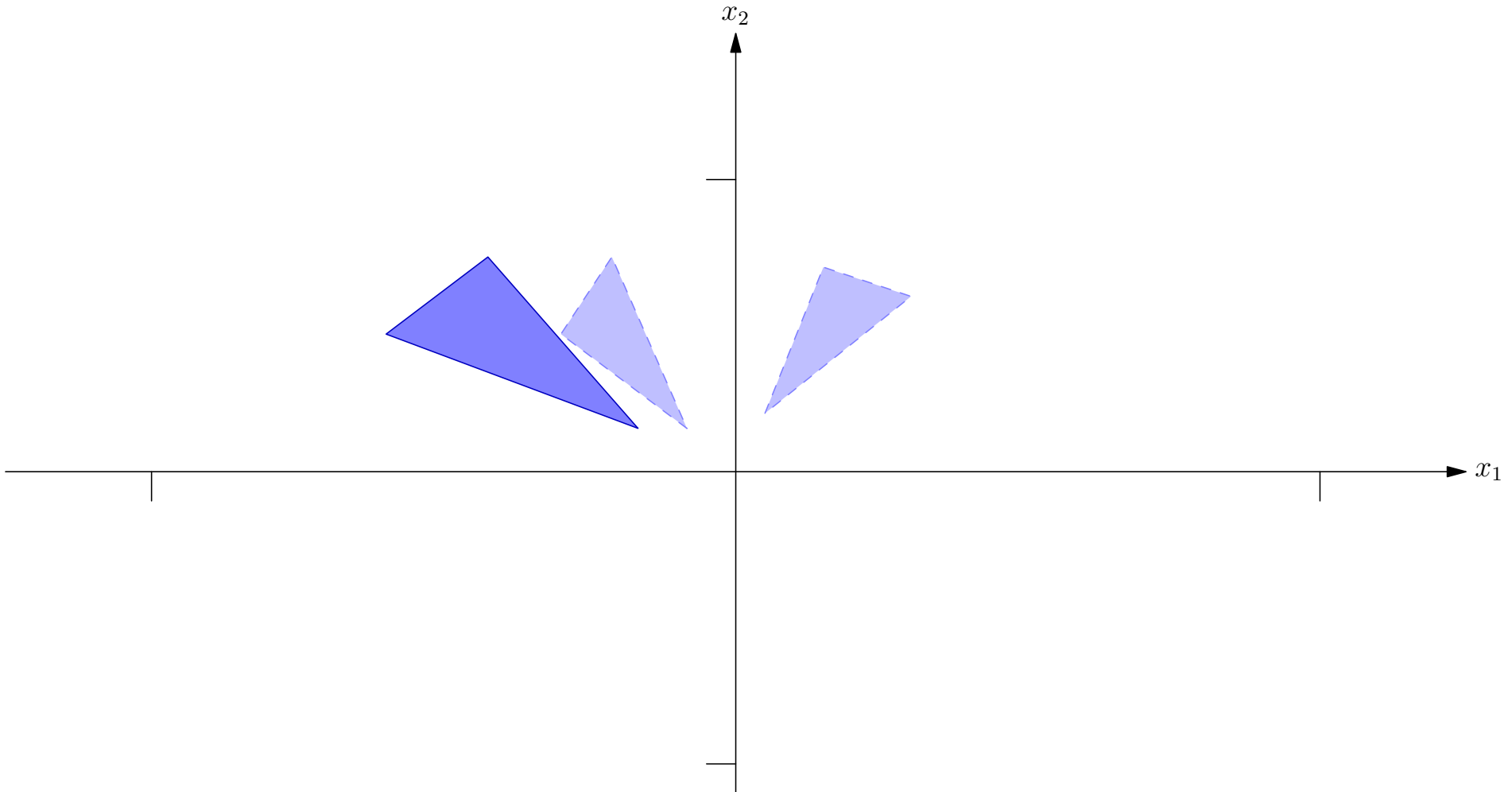
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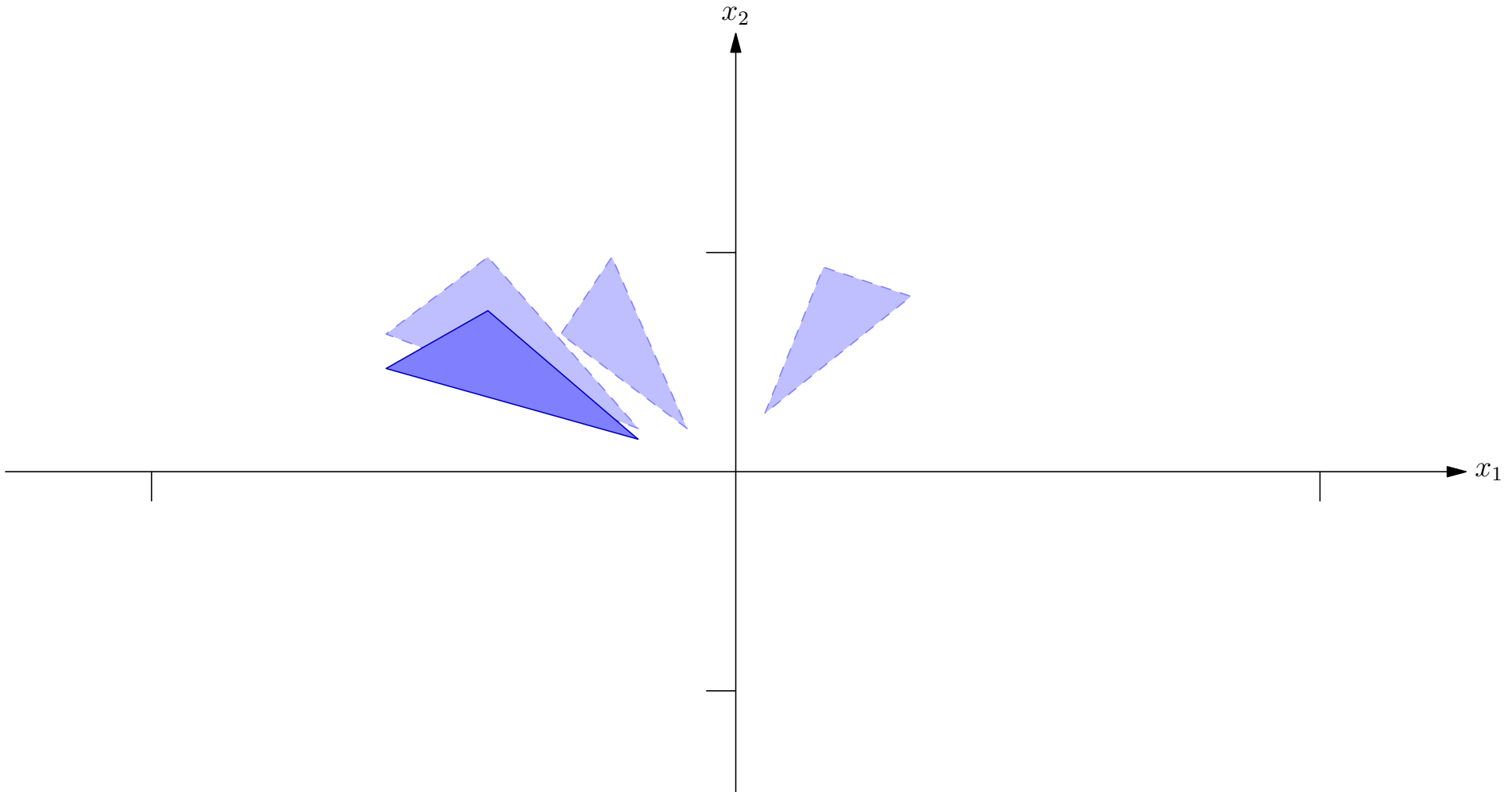
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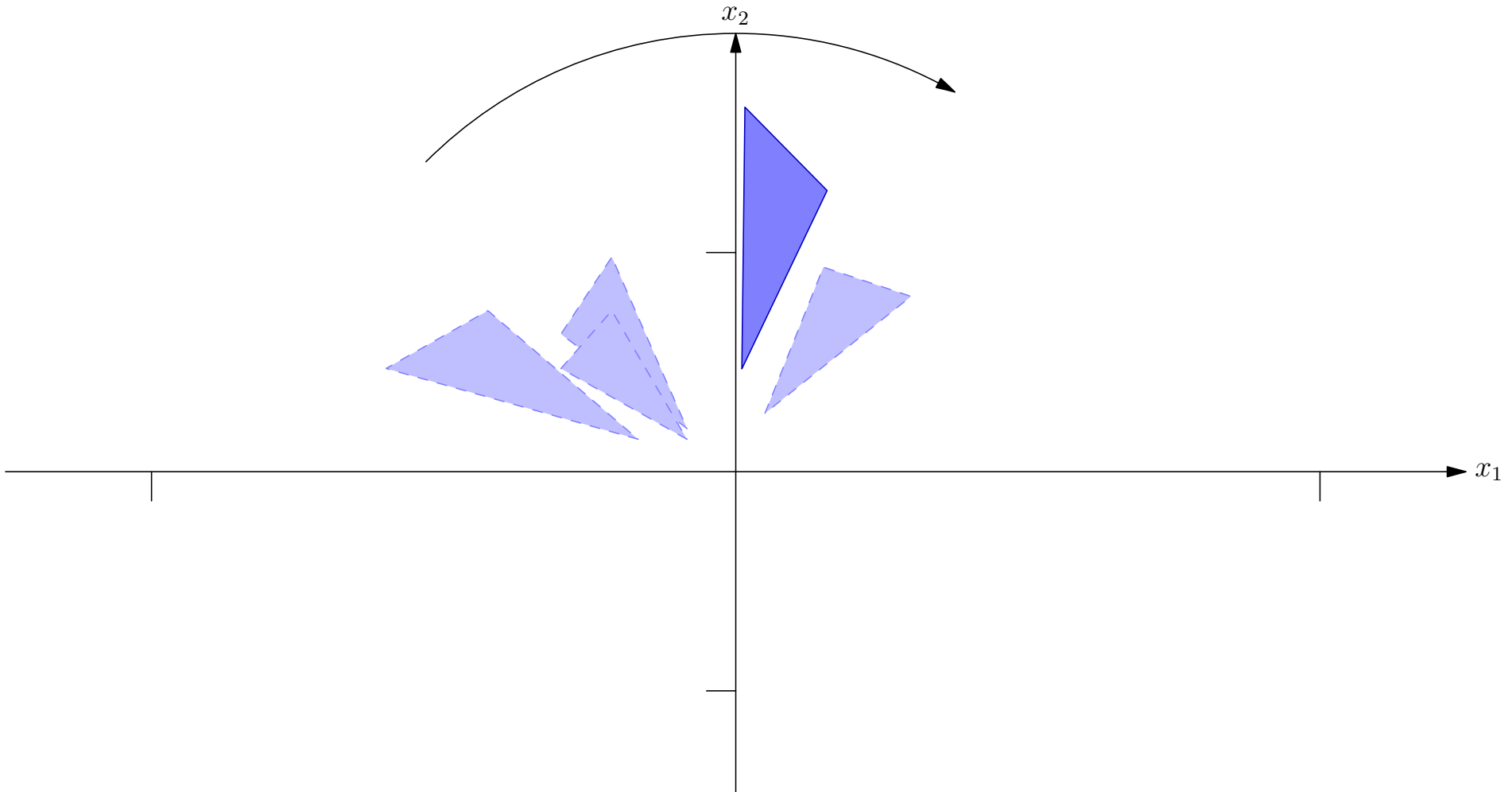
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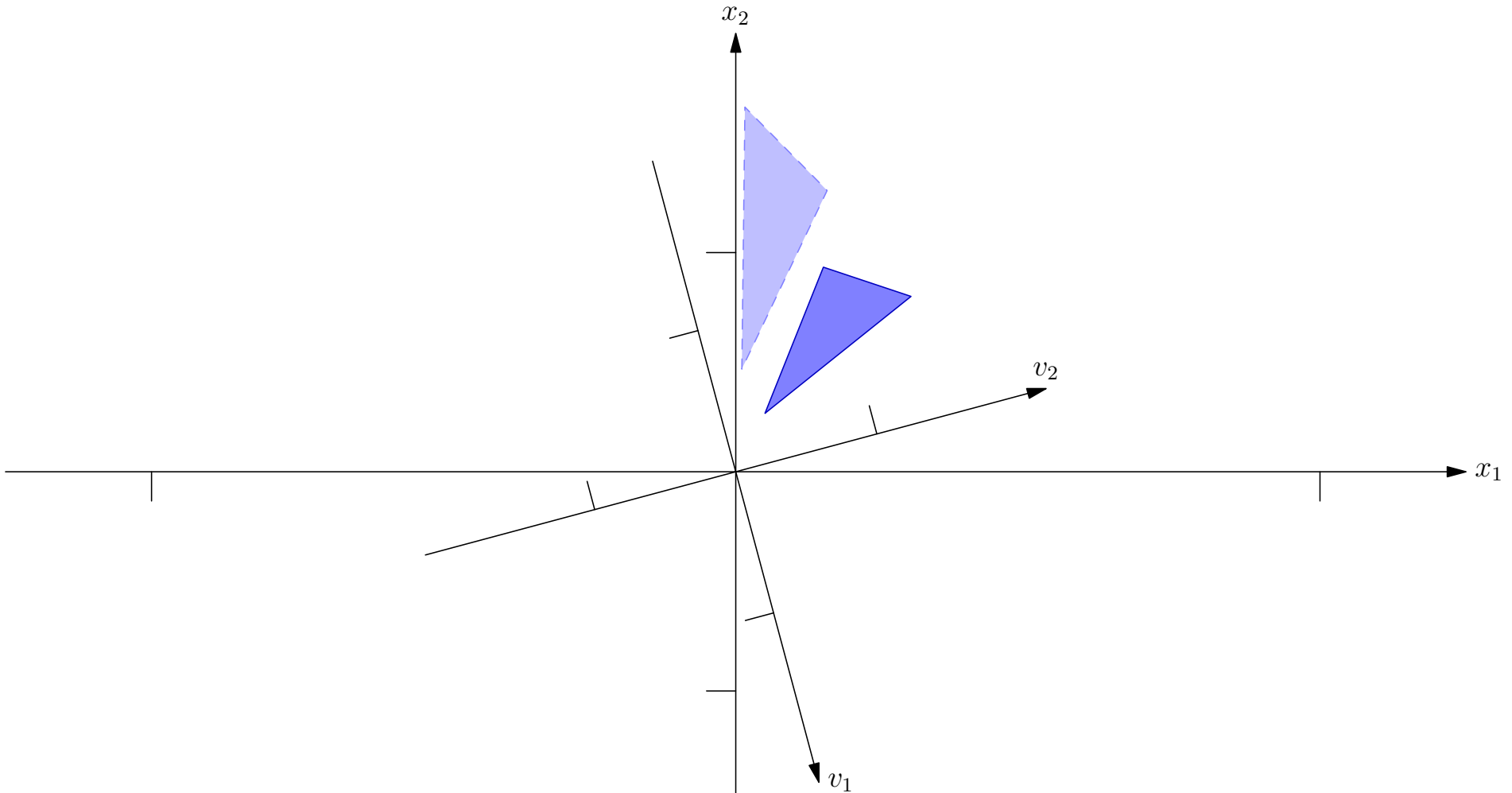
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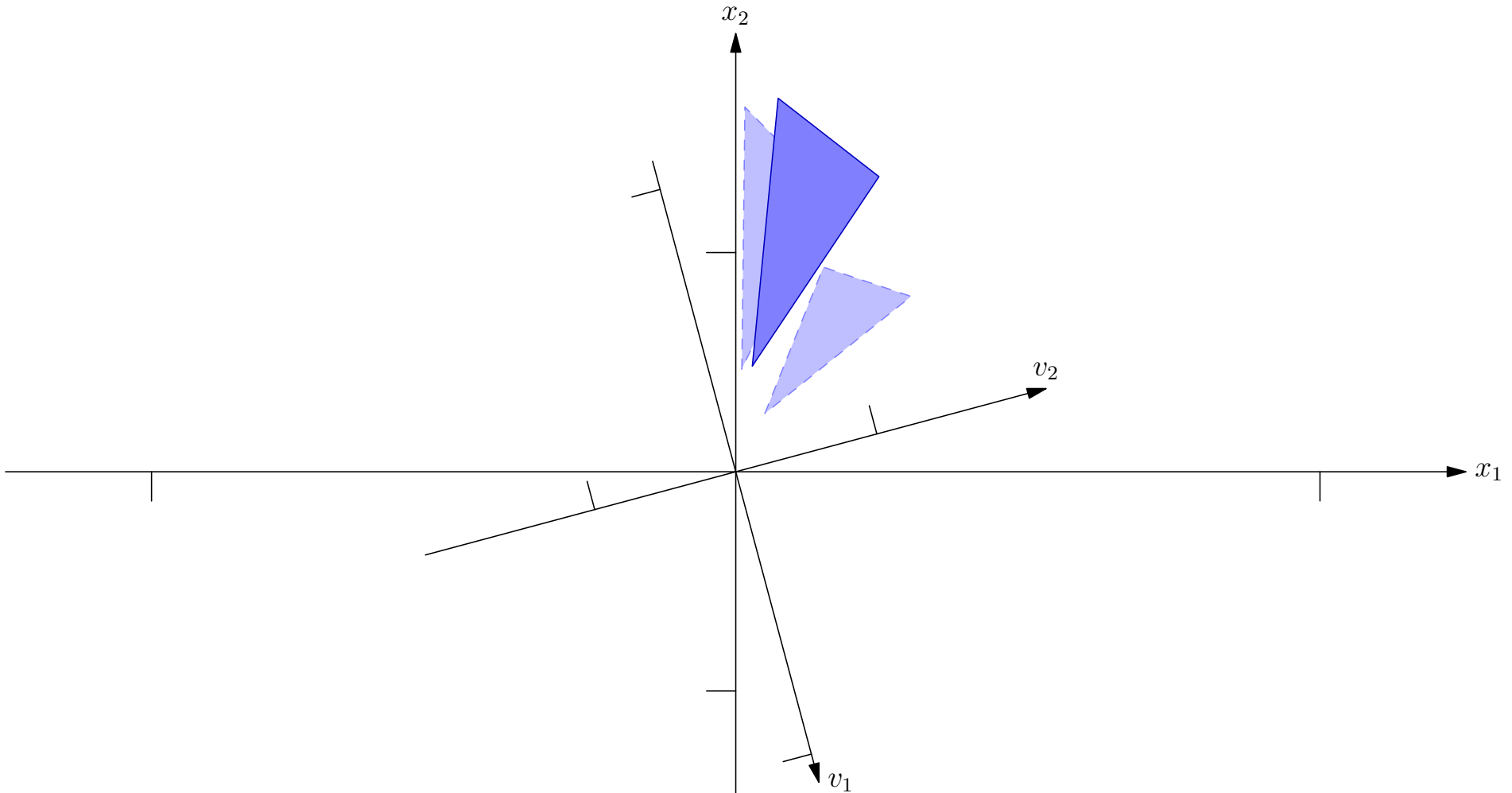
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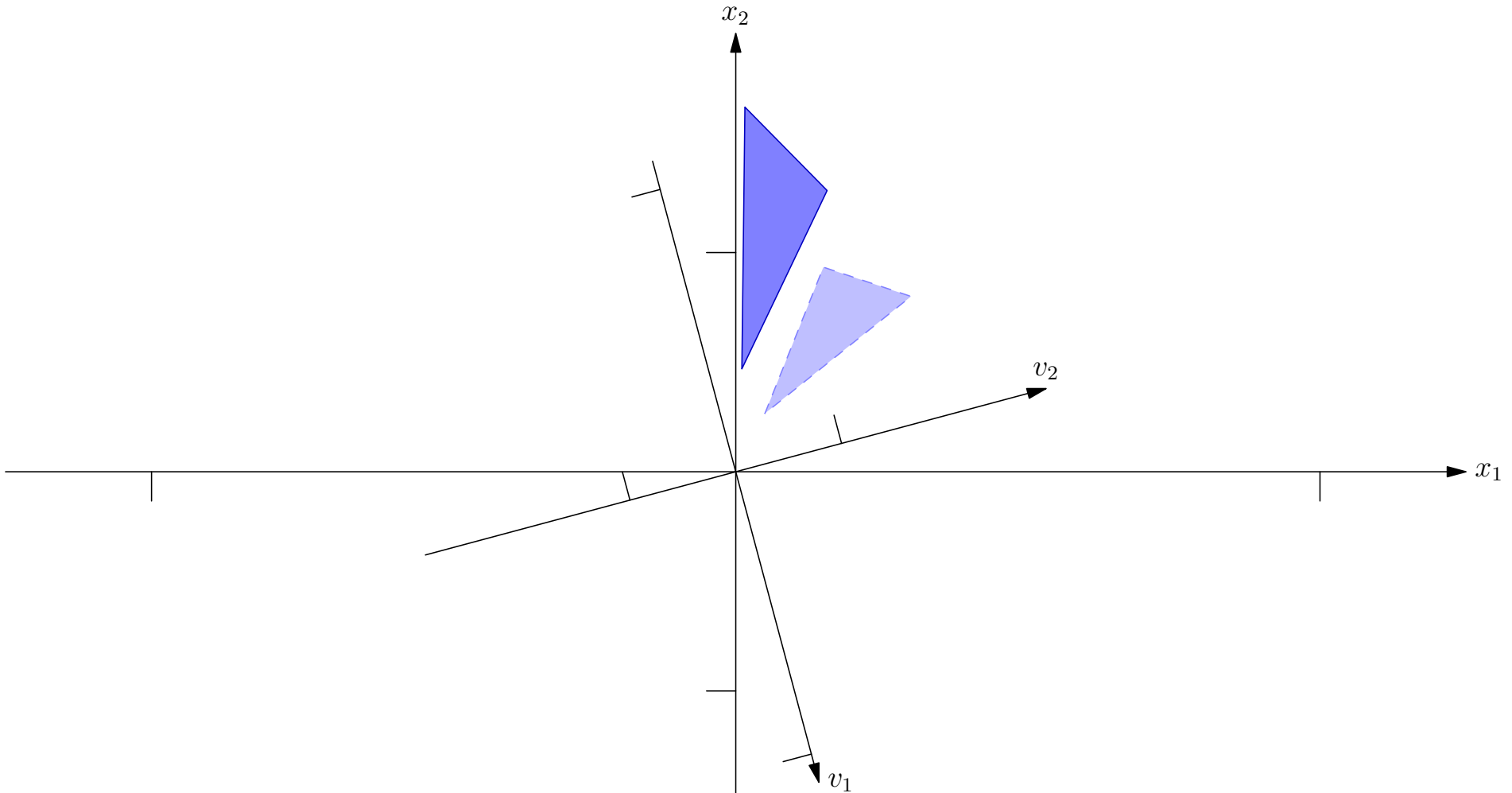
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# Inverses

- For any symmetric invertible matrix

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

$$\mathbf{M}^{-1} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T$$

- Where  $\mathbf{\Lambda}^{-1} = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\lambda_n} \end{pmatrix}$

- Since

$$\begin{aligned} \mathbf{M}\mathbf{M}^{-1} &= (\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T)(\mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T) = \mathbf{V}\mathbf{\Lambda}(\mathbf{V}^T\mathbf{V})\mathbf{\Lambda}^{-1}\mathbf{V}^T \\ &= \mathbf{V}\mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{V}^T = \mathbf{V}\mathbf{V}^T = \mathbf{I} \end{aligned}$$

- I.e, Small eigenvalues become large eigenvalues and visa verse

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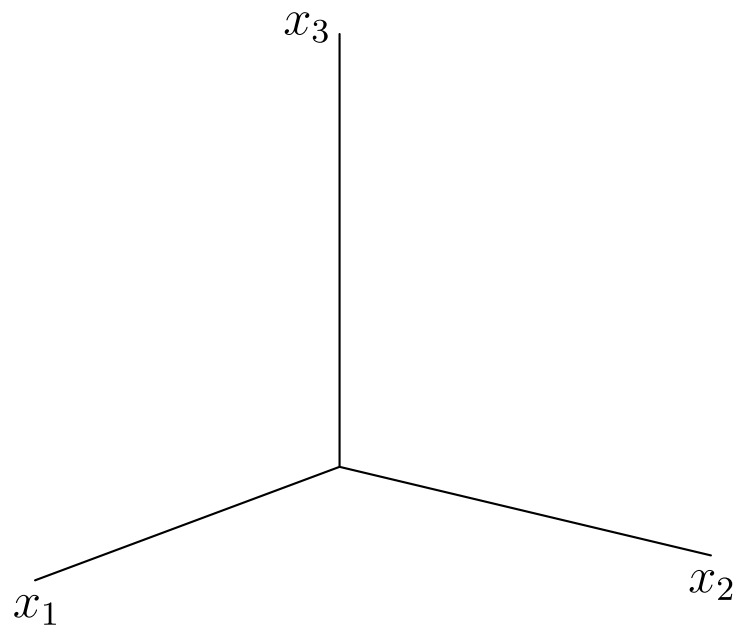
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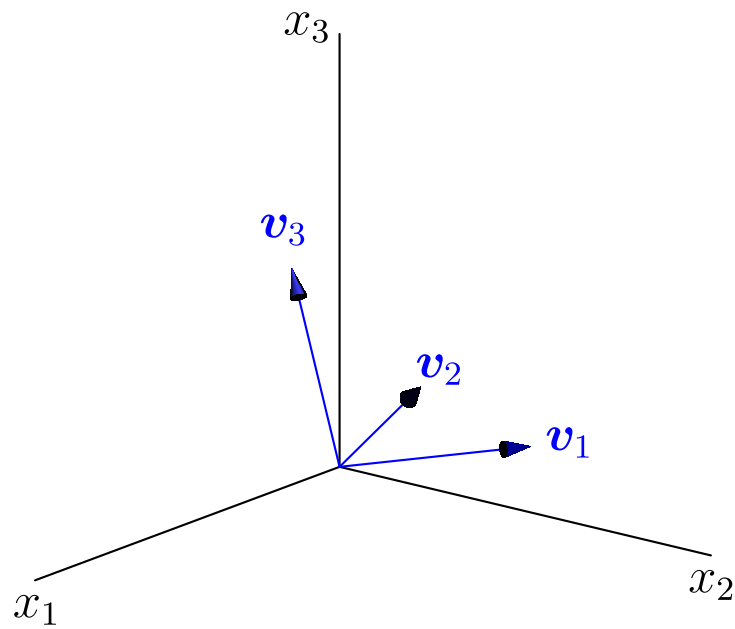
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# III-Conditioning Again



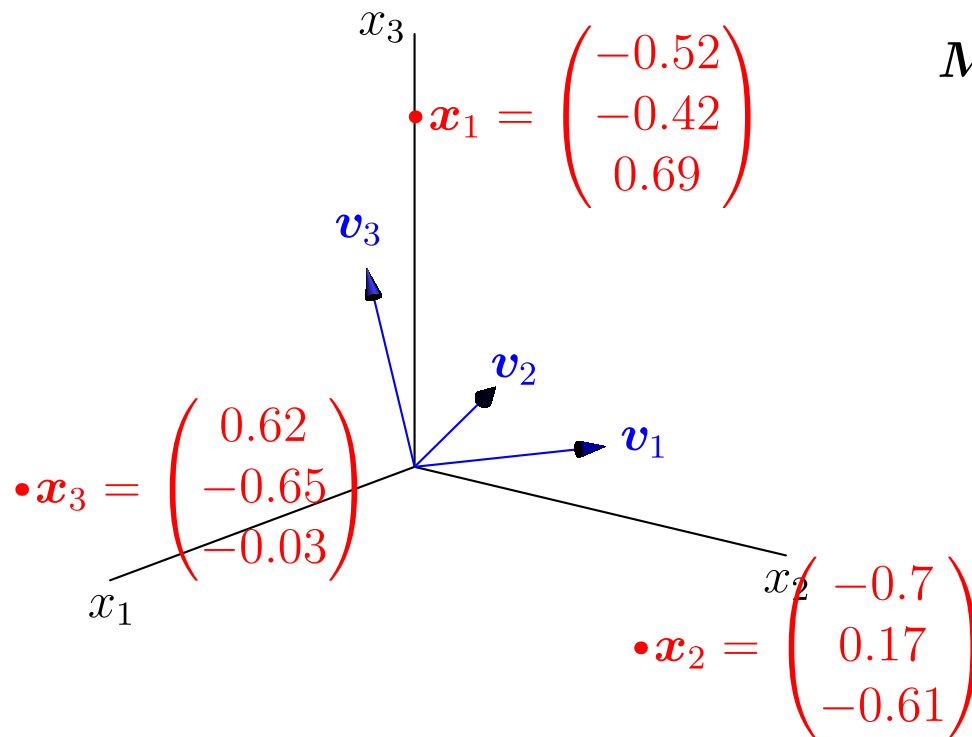
$$\begin{aligned} \mathbf{M} &= \begin{pmatrix} 0.80 & 0.01 & 0.25 \\ 0.01 & 0.85 & 0.48 \\ 0.25 & 0.48 & 0.5 \end{pmatrix} \\ &= \mathbf{V} \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 1.25 & 0 \\ 0 & 0 & 0.1 \end{pmatrix} \mathbf{V}^T \end{aligned}$$

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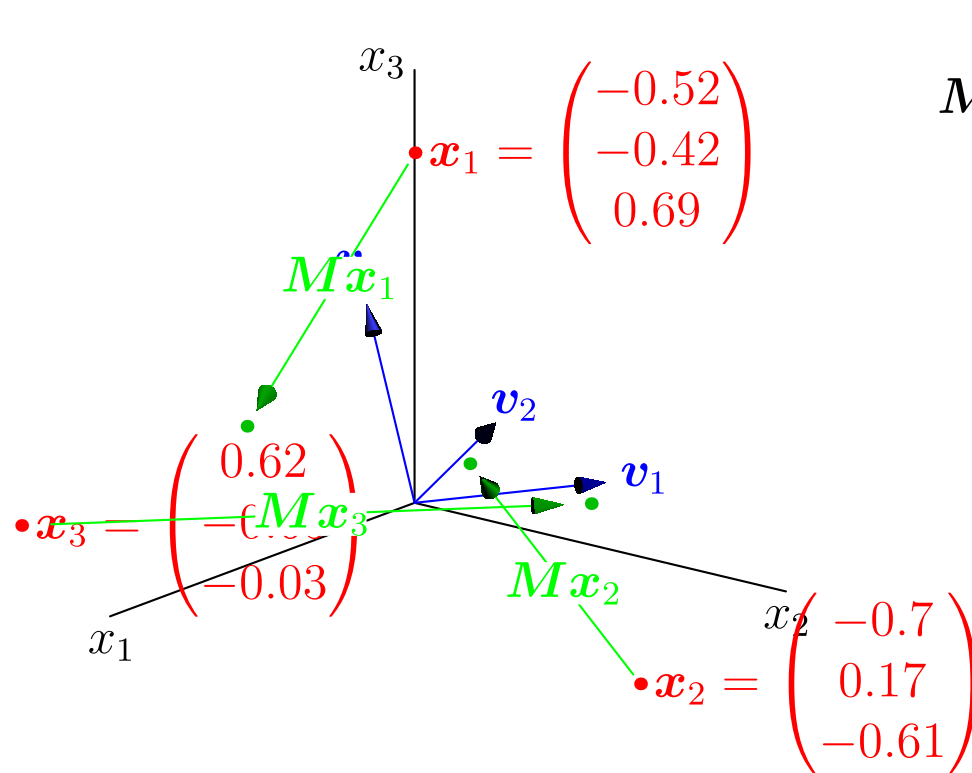
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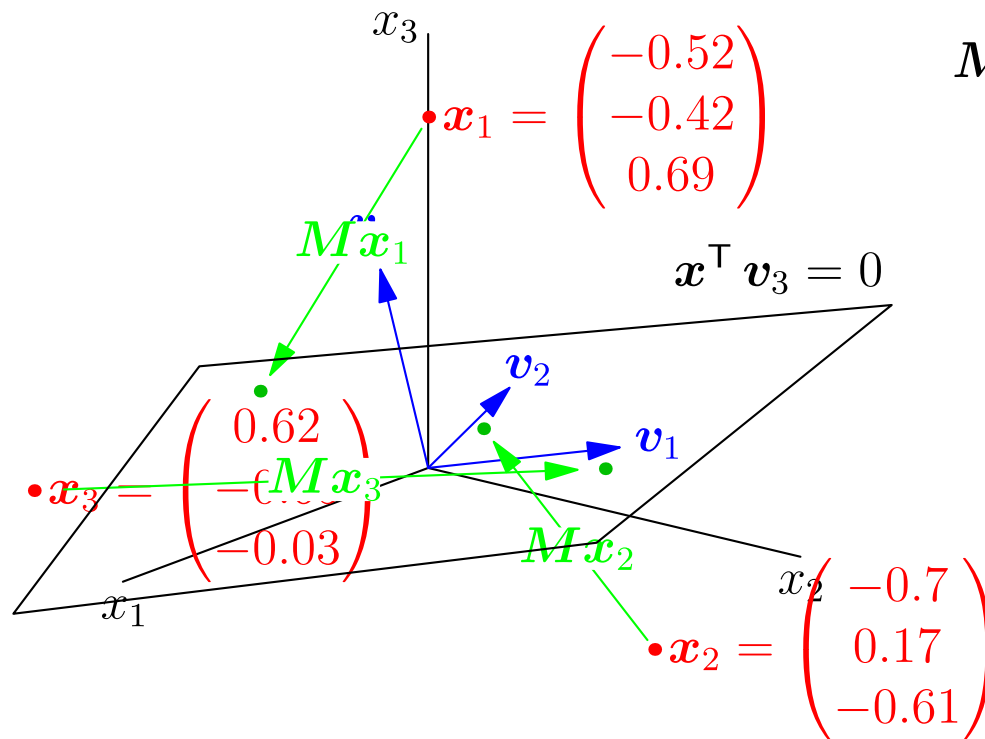
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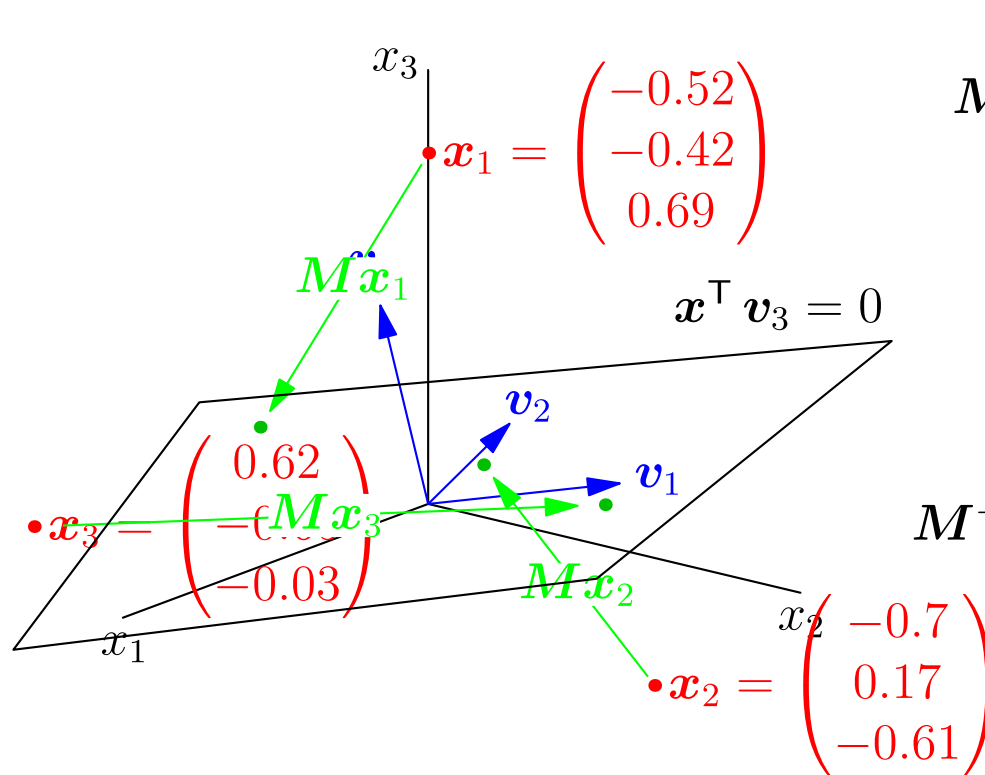
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$$M^{-1} = \begin{pmatrix} 1.9 & 1.17 & -2.1 \\ 1.17 & 3.34 & -3.8 \\ -2.1 & -3.8 & 6.8 \end{pmatrix}$$

$$= V \begin{pmatrix} 1.25 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 10 \end{pmatrix} V^T$$

# Condition Number

- Taking matrix inverses can be inherently unstable
- Any small error can be amplified by taking the inverse
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)
- For invertible matrices we can take the largest eigenvalue as a norm of the matrix
- The condition number is given by

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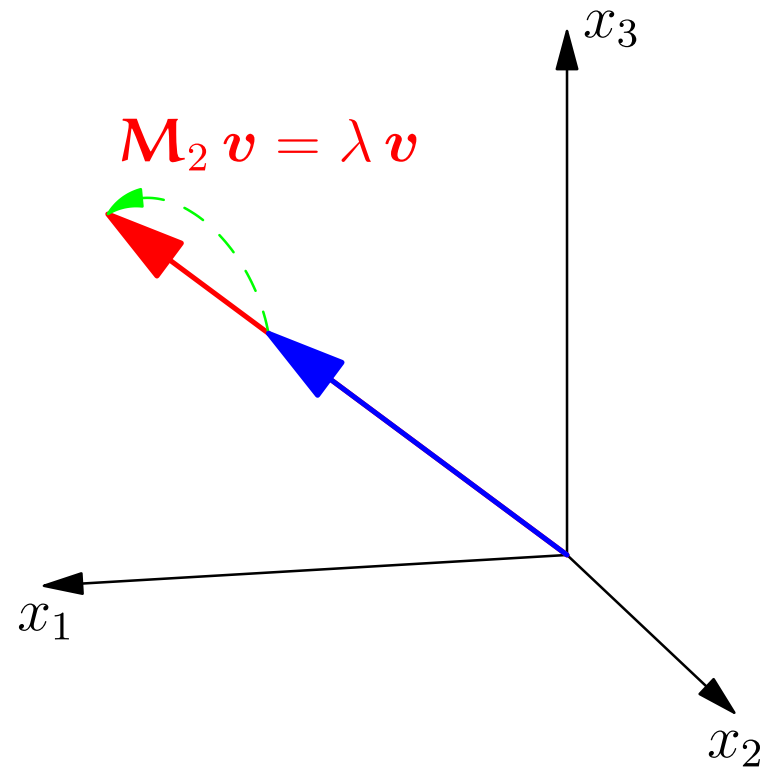
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# Outline

1. Eigenvectors
2. Orthogonal Matrices
3. Eigen Decomposition
4. **Low Rank Approximation**



# Rank of a Matrix

- The rank of a matrix,  $\mathbf{M}$ , is the number of non-zero eigenvalues
- The space spanned by the eigenvectors  $\mathbf{v}_a$ ,  $\mathbf{v}_b$ , etc. with zero eigenvalue forms a **null space**
- Any vector in the null space will get projected to the zero vector

$$\mathbf{M}(a\mathbf{v}_a + b\mathbf{v}_b + \dots) = \mathbf{0}$$

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- Recall that matrices with large and small eigenvalues are ill-conditions so the inverse has the potential to greatly amplify any measurement error
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# Summary

- Linear mappings are commonly used in machine learning algorithms such as regression
- We can understand symmetric operators by looking at their eigenvectors
- Any symmetric matrix can be decomposed as  $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$ 
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